

KUALANAMU MEDAN, Indonesia

WMO: 960350

Lat: 3.640N

Lon: 98.879E

Elev: 82

StdP: 14.65

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	72.7	73.2	68.4	104.5	84.4	69.7	109.4	82.4	14.4	84.0	13.1	84.5	1.6	240	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	14.3	93.5	79.0	92.6	79.1	91.6	79.0	82.1	88.4	81.5	87.9	80.9	87.5	8.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.7	160.0	85.3	79.9	155.5	84.8	79.1	151.5	84.3	46.1	88.5	45.3	88.2	44.7	87.9	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.2	12.3	11.3	DB	70.0	96.7	1.5	1.8	69.0	98.0	68.1	99.1	67.3	100.1	66.3	101.4	
			WB	68.3	83.6	1.4	1.4	67.3	84.6	66.5	85.4	65.7	86.2	64.7	87.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.1	80.9	81.6	82.6	83.1	83.4	83.2	82.7	82.2	81.7	81.4	81.2	80.7
	DBStd	1.77	1.50	1.51	1.46	1.57	1.61	1.62	1.58	1.63	1.50	1.49	1.31	1.52
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11703	959	886	1011	993	1034	996	1013	999	951	973	936	952
	CDD65	6228	494	466	546	543	569	546	548	534	501	508	486	487
	CDH74	62743	4635	4669	5801	5905	6193	5957	5749	5374	4726	4734	4552	4448
CDH80	23468	1478	1687	2339	2409	2538	2496	2361	2108	1731	1580	1430	1311	
Wind	WSAvg	3.7	3.8	3.8	3.9	3.5	3.6	3.7	3.8	3.8	3.7	3.5	3.6	3.8
Precipitation	PrecAvg	100.4	6.2	3.9	5.9	5.8	8.7	6.7	6.9	8.9	12.5	13.4	10.6	10.7
	PrecMax	147.5	14.5	11.4	15.2	13.5	17.8	14.6	12.3	17.6	22.5	25.0	20.0	20.2
	PrecMin	64.4	2.8	1.6	1.3	0.9	3.1	3.1	2.3	4.1	6.7	7.4	4.0	4.3
	PrecStd	20.6	2.8	2.3	3.8	3.1	3.8	2.7	2.7	3.3	3.7	3.7	3.5	4.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.6	91.8	93.5	94.1	94.9	95.1	94.6	93.9	92.5	91.7	91.1	90.0
		MCWB	78.5	78.8	79.1	80.0	79.3	77.8	77.8	78.3	78.8	78.4	78.4	78.3
	2%	DB	89.3	90.5	91.7	92.3	93.0	92.9	92.6	91.7	90.9	89.8	89.4	88.4
		MCWB	78.1	78.4	78.8	79.7	79.8	78.9	78.3	78.6	78.6	78.4	78.4	78.3
	5%	DB	88.0	89.4	90.7	91.2	91.5	91.4	91.2	90.3	89.5	88.4	87.9	87.6
		MCWB	78.2	78.2	78.8	79.5	79.8	79.2	78.6	78.6	78.4	78.4	78.5	78.2
	10%	DB	86.6	87.9	89.4	89.7	89.8	89.9	89.6	89.1	87.9	87.4	86.5	86.1
		MCWB	77.9	78.1	78.7	79.2	79.5	79.1	78.5	78.6	78.6	78.4	78.4	78.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.9	81.0	81.9	82.4	82.8	82.8	82.3	82.3	81.7	81.6	81.3	81.0
		MCDB	87.1	88.1	88.8	89.4	89.5	89.0	87.8	88.5	87.1	86.8	85.8	86.0
	2%	WB	79.9	80.0	80.8	81.4	81.8	81.6	81.0	80.8	80.4	80.1	80.1	80.0
		MCDB	85.9	86.8	88.0	88.6	88.5	88.1	87.3	87.5	86.9	86.0	85.6	85.2
	5%	WB	79.1	79.4	80.1	80.7	81.0	80.7	80.1	79.9	79.6	79.4	79.5	79.2
		MCDB	85.0	86.0	87.2	88.0	87.8	87.5	86.8	86.7	86.2	85.2	85.0	84.4
	10%	WB	78.6	78.9	79.5	80.0	80.2	79.9	79.4	79.1	79.0	78.9	78.9	78.7
		MCDB	84.3	85.3	86.4	87.0	86.9	86.7	86.2	86.0	85.5	84.7	84.4	83.8
Mean Daily Temperature Range	5% DB	MDBR	12.6	13.8	14.7	14.4	14.3	14.9	15.1	14.6	13.8	12.7	12.1	11.7
		MCDBR	14.3	15.7	16.2	16.0	16.2	16.6	17.2	16.5	15.7	14.6	13.8	13.6
		MCWBR	5.7	6.0	6.1	6.1	6.2	6.4	6.5	6.4	6.2	5.7	5.6	5.5
	5% WB	MCDBR	13.0	14.1	14.8	14.7	14.5	15.0	15.3	15.1	14.2	13.0	12.4	12.0
		MCWBR	5.8	6.0	6.3	6.2	6.3	6.5	6.6	6.5	6.4	5.8	5.6	5.5
Clear-Sky Solar Irradiance	taub	0.477	0.524	0.504	0.473	0.466	0.479	0.469	0.489	0.488	0.498	0.453	0.453	
	taud	2.255	2.102	2.175	2.272	2.309	2.274	2.303	2.231	2.235	2.203	2.342	2.340	
	Ebn at Noon	270	259	264	267	262	255	259	259	264	263	275	274	
	Edh at Noon	45	53	49	44	41	42	41	45	46	48	41	41	
All-Sky Solar Radiation	RadAvg	1403	1549	1618	1540	1513	1542	1504	1505	1442	1379	1306	1286	
	RadStd	83	80	87	67	87	88	77	105	113	155	60	89	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (1 neighbor)	+0.34	+0.20	N/A	N/A	+0.72	+1.01	N/A	N/A	N/A	N/A